16_proprioceptors_and_reflexes

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16. Proprioceptors and Reflexes

Proprioceptors

- **Proprioception:** positional sense of the body/joint in three-dimensional space.
- **Kinesthesia:** sense of body/joint motion.
- Proprioceptors/kinesthesia receptors listed in the source:
    - **Muscle spindle:** senses lengthening of muscle.
    - **Golgi tendon organ/apparatus(GTO):** senses muscle-tendon tension.
    - **Joint-capsule receptors:** slow-adapting receptors that give tonic sensation of joint position.

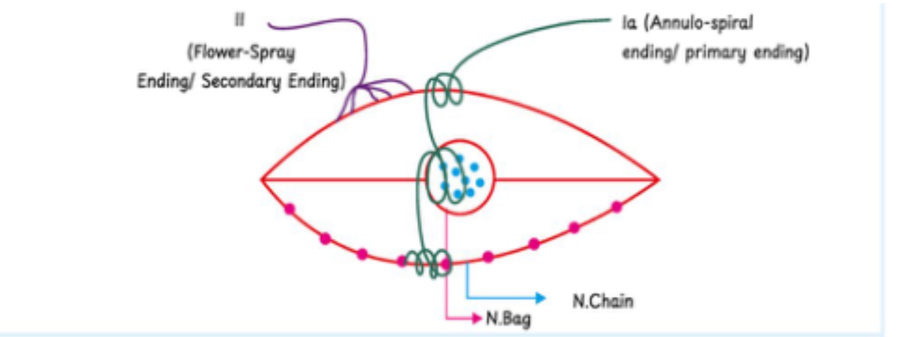
Muscle spindle

Location and structure

Afferent and efferent connections of the muscle spindle

Ia afferent fibre

- Myelinated fibre originating from the central region of the muscle spindle.
- Originates from all fibre types: nuclear bag and nuclear chain fibres.
- Detects both velocity and an increase in muscle length (wording retained from source).
- Makes a spiral synapse around the muscle spindle, called the**annulo-spiral ending/primary ending**.

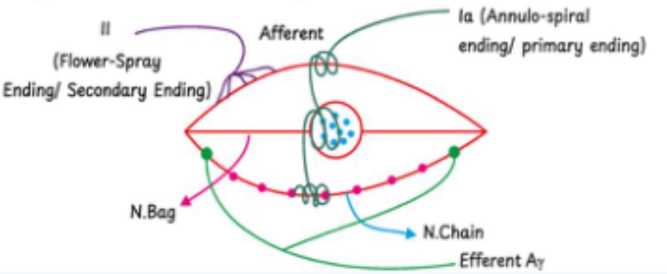


"IIa type of fibre" (source wording)

- Origin: adjacent to the centre of the muscle spindle.
- Makes a flower-spray/secondary ending.
- Detects only an increase in muscle length.
- Connects static fibres of both nuclear bag and nuclear chain fibres.

Efferent muscle fibre

- γ(gamma)fibre supplies the most terminal portion of the muscle spindle.

	○ Connects static fibres of both nuclear bag fibres & nuclear chain fibres
EFFERENT MUSCLE FIBRE	• Aγ fibre supplies the most terminal portion of the muscle spindle 

- The source states that only two receptors in the body receive efferent supply from the CNS:
    - Muscle spindle.
    - Inner hair cell(organ of Corti)/outer hair cell.

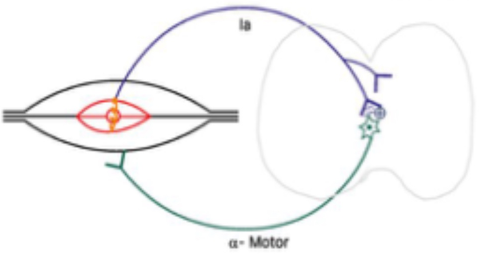
Stretch reflex / myotactic reflex

Pathway

- **Receptor:** muscle spindle.
- **Afferent:** Ia fibre activates, reaches the spinal cord and makes a direct synapse with a motor neuron supplying the same muscle's extrafusal muscle fibres.
- **Centre:** spinal cord.
- **Efferent:** α motor neuron.
- The muscle spindle detects increased muscle length.

Sequence:

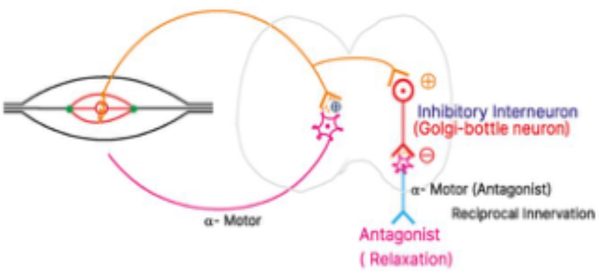
1. Stretching the muscle → muscle spindle generates a receptor potential.
2. Action potential is generated on the Ia fibre.
3. Ia fibre produces/releases glutamate at its synapse with the motor neuron.
4. The motor neuron is stimulated → extrafusal fibres are stimulated → skeletal-muscle contraction.



- Stretch reflex:sudden/passive stretch of a skeletal muscle leads to reflex contraction.
- Examples of deep tendon reflexes:biceps jerk,triceps jerk,knee jerk and ankle jerk; catching a ball is also listed.
- Active stretch is suppressed by the cerebral cortex.
- Number of synapses: **one(monosynaptic pathway)** .
- Reaction time for skeletal-muscle contraction following stretch of the muscle spindle: **19–24 ms**;synaptic delay: **0.5 ms**.

Reciprocal innervation pathway / Sherrington's law

- The Ia fibre gives a branch at the spinal-cord level → stimulates an inhibitory interneuron(called “Golgi-bottle neuron” in the source)→ acts on the α motor neuron of the antagonist muscle and inhibits it → antagonist muscle relaxes.
- Number of synapses: **two(bisynaptic pathway)** .



Role of the γ motor neuron

- Present in the anterior horn of the spinal cord.
- γ motor neuron supplies the terminal end of the intrafusal fibre/muscle spindle.

- Actin and myosin are present only at the terminal portion of the muscle spindle.

Sequence shown in the source:

- Stimulation of the A-gamma efferent neuron → contraction of the two terminal ends of the muscle spindle → increased length of its central portion → stimulation of the Ia fibre → stimulation of the alpha motor neuron → muscle contraction.
- γ motor neuron controls the sensitivity of the muscle spindle/stretch reflex.
- Jendrassik manoeuvre,anxiety,sympathetic stimulation,severe pain and fever are listed as stimuli of γ motor activity that exaggerate muscle contraction/reflexes.
- Jendrassik manoeuvre(clenching teeth/interlocking fingers)is performed when the deep tendon reflex is not elicited.

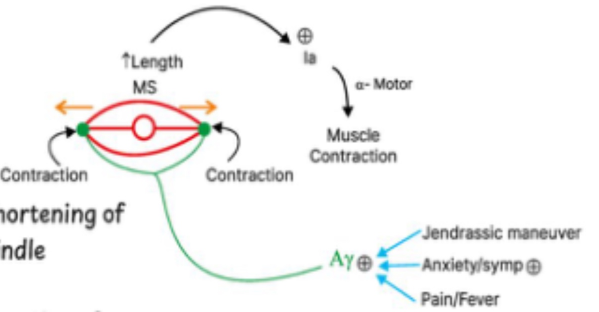
α–γ co-activation

- Jendrassik manoeuvre activates the motor cortex; the fibre reaching the spinal cord stimulates both α motor and γ motor neurons. This is called**α–γ co-activation**.
- α motor-neuron stimulation → muscle contraction and shortening → the spindle would become inactive(described in the source as an unacceptable condition).
- The muscle spindle should remain active during contraction because it is responsible for the stretch reflex that maintains posture.
- γ motor-neuron stimulation increases muscle-spindle length, allowing the spindle to remain active.

- Jendrassic maneuver (clenching of teeth, interlocking of fingers) is performed when DTR is not elicited

α–γ CO-ACTIVATION

- Jendrassic maneuver → activates motor cortex → fibre reach at level of spinal cord → stimulate both α motor neuron and Aγ motor neuron
  - This is k/as α–γ co-activation
- α motor neuron stimulation → muscle contraction → hortening of muscle → ↓ muscle spindle length inactive muscle spindle (unacceptable condition)





Golgi tendon organ and inverse stretch reflex

- A single GTO contains **3–25 fibres**, average **10–15 fibres**.
- Activation: increased tension at the muscle tendon during active contraction or passive stretching.
- Afferent from GTO: myelinated **Ib fibres**.

Inverse stretch reflex / autogenic inhibition

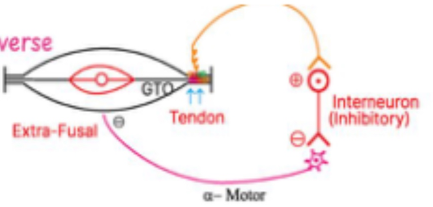
Sequence:

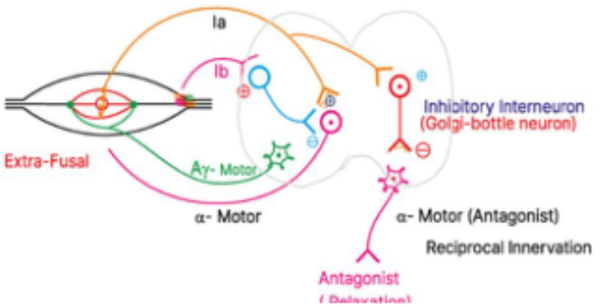
1. Sudden skeletal-muscle contraction → increased tension at the muscle tendon.
2. GTO activation → signal from the Ib neuron.
3. Ib signal activates inhibitory interneurons, which inhibit the α motor neuron.
4. Muscle fibre relaxes → tendon tension is reduced, helping prevent tendon rupture.

- Sudden contraction of skeletal muscle leads to reflex relaxation: inverse stretch reflex/autogenic inhibition.
- Protective response.

- **Receptor:** GTO; **afferent:** Ib fibre; **synapses:** two(bisynaptic pathway).

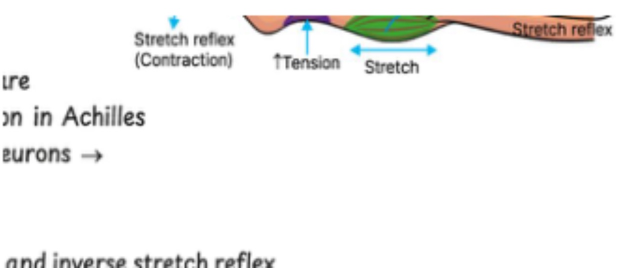
- Sudden contraction of skeletal muscle leads to reflex relaxation → inverse stretch reflex or autogenic inhibition
- Protective response
- Receptor for inverse stretch reflex: GTO
- Afferent: Ib fibre
- Synapse: 2 i.e., Bisynaptic





Clonus: ankle clonus

- **Clonus:** rhythmic alternating contraction and relaxation of a muscle group.
- Seen in an upper motor neuron(UMN)lesion.
- The source states that the stretch reflex and inverse stretch reflex are exaggerated because of loss of the brain's inhibitory effect on these reflexes.
- In UMN lesion, dorsiflexion of the foot passively stretches the gastrocnemius → stretch reflex → gastrocnemius contracts → plantar flexion.
- Opposing plantar flexion by dorsiflexion increases Achilles-tendon tension → inverse stretch reflex inhibits the α motor neuron through interneurons → gastrocnemius relaxes.
- Clonus is described as sequential activation of the stretch reflex and inverse stretch reflex.

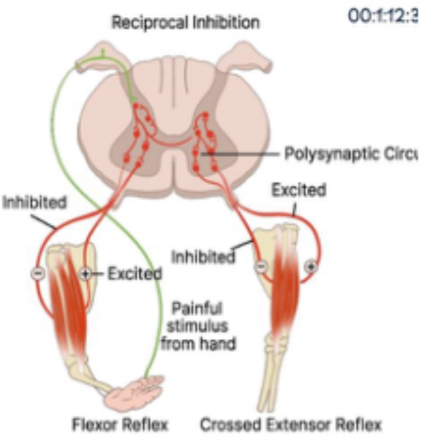


Withdrawal reflex

- A protective reflex.
- A painful stimulus from the hand is carried to the spinal cord through pain/temperature fibres. Multiple spinal-cord interneurons cause excitation of the flexor muscle group and inhibition of the extensor muscle group. This is reciprocal innervation.
- On the opposite side of the body, the extensor group is excited and the flexor group inhibited → **crossed extensor reflex**.
- Typical polysynaptic reflex.
- Afferents listed in the source:group III and IV fibres for painful stimuli;group II fibres for pressure.

Components

- Flexor reflex: withdrawal of the limb from the stimulating object.
- Reciprocal inhibition: relaxation of antagonist muscles.
- Crossed extensor reflex: occurs 0.2–0.5 seconds after the stimulus.
- After-discharge phenomenon: firing of excited interneurons lasts for many fractions of a second after the stimulus ends.



MCQs

1. Golgi tendon organ of skeletal muscle is stimulated by:

- a. Tension in the muscle
- b. Length of the muscle
- c. Velocity of muscle
- d. Muscle position

Answer: (a)

2. Golgi tendon organ senses which of the following?

- a. Tension
- b. Muscle stretch
- c. Velocity of muscle
- d. Pain during stretch

Answer: (a)